JOURNAL OF CHEMICAL EDUCATION

Open Access

This article is licensed under CC-BY 4.0

pubs.acs.org/jchemeduc

Article

Development of a Teaching Approach for Structure Elucidation Using 1D and 2D Homonuclear and Heteronuclear NMR Spectra

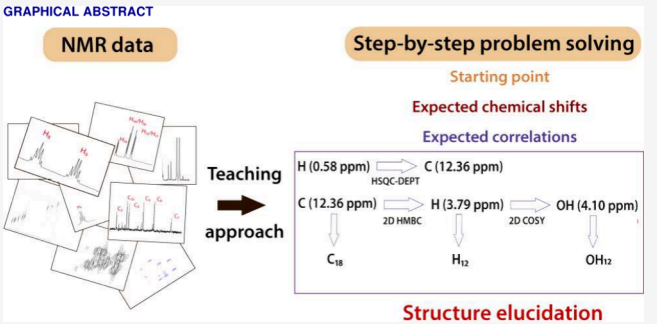
Eleni Chontzopoulou, Andromachi Tzani, Katerina Paschalidou, N. Zoupanou, and Thomas Mavromoustakos*

Cite This: J. Chem. Educ. 2025, 102, 3226−3233

Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

ABSTRACT: A frequent challenge that graduate and undergraduate students in chemistry, biology, and pharmacy laboratories face is accurately assigning proton and carbon peaks in the 1D and 2D Nuclear Magnetic Resonance (NMR) spectra of organic and pharmaceutical molecules. We propose a consistent, step-by-step approach to effectively assist students in simultaneously interpreting a variety of 1D and 2D NMR spectra. According to this approach, the anticipated NMR peaks are initially predicted based on the compound's molecular structure, followed by the interpretation of the actual spectra. The novelty of this teaching approach lies in the simultaneous use of experimental data from 1D and 2D homonuclear and heteronuclear spectra to validate or enhance the theoretical approach derived solely from the molecule's two-dimensional structure. The integrations obtained from 1D ^{1}H NMR spectrum and bond and spatial correlations from homonuclear and heteronuclear 2D NMR spectra are the experimental data used for every structure elucidation. A special notation using arrows is adduced in order to indicate the bond and spatial correlations and to schematically present the proposed teaching approach in an easy and comprehensive way. The proposed notation could offer a new way to visualize and conceptualize the theoretical approach, potentially making it easier for students to apply. This methodology begins with predicting proton, carbon, and proton-carbon correlations based solely on the molecular structure, followed by listing the expected signals for all available spectra. The approach then emphasizes comparing these predicted correlations and signals with those observed in 1D and 2D NMR spectra. This methodology is only applicable when the synthesis of a particular product is anticipated, which is a typical scenario for students engaged in organic synthesis laboratories or conducting research in organic and medicinal chemistry.



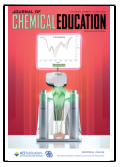
KEYWORDS: *NMR spectroscopy, NMR teaching approach, structure elucidation, organic compounds*

INTRODUCTION

NMR spectroscopy has become an essential analytical tool for structure elucidation in various types of chemical laboratories. The advancements in NMR spectroscopy nowadays enable the acquisition of 1D ^{1}H, ^{13}C and 2D COSY, NOESY, HSQC, and HMBC spectra in under 8 h. As a result, the routine handling of a variety of homonuclear and heteronuclear 1D and 2D NMR spectra for structure elucidation has become a part of daily workflows in chemical laboratories. Thus, future scientists must be trained not only in the theoretical principles of NMR but also in the integrated use of various NMR spectra for structure determination.

Undergraduate students should be familiar with these techniques and fluent in interpreting different kinds of NMR spectra. Hence, NMR courses have been included in the curriculum of different degrees in universities worldwide related to Pharmacy, Chemistry, Material and Biology studies. One of the optional courses in the Department of Chemistry of the National and Kapodistrian University of Athens (NKUA) is "NMR Spectroscopy and Molecular Modeling", which has been taught for the last 10 years and focuses on the NMR interpretation of small organic molecules. To complete this course, students are required to assign the proton and carbon peaks of known molecular structures using all of the necessary 1D and 2D NMR spectra. This exercise focuses on peak assignment rather than structure elucidation. Instructors have observed that students struggle to assign peaks in complex compounds, especially when interpreting multiple spectra during homework exercises or exams. This phenomenon seems to be a little paradoxical since, when possessing all the necessary spectra, it is quite easier to assign proton and carbon

Received: September 5, 2024
Revised: June 25, 2025
Accepted: June 27, 2025
Published: July 9, 2025



ACS Publications

© 2025 The Authors. Published by American Chemical Society and Division of Chemical Education, Inc.

https://doi.org/10.1021/acs.jchemed.4c00402
J. Chem. Educ. 2025, 102, 3226−3233

Downloaded from pubs.acs.org/jceda8/article-pdf/102/8/3226/4129173/ed4c00402.pdf by Dong Ziyu on 10 October 2026

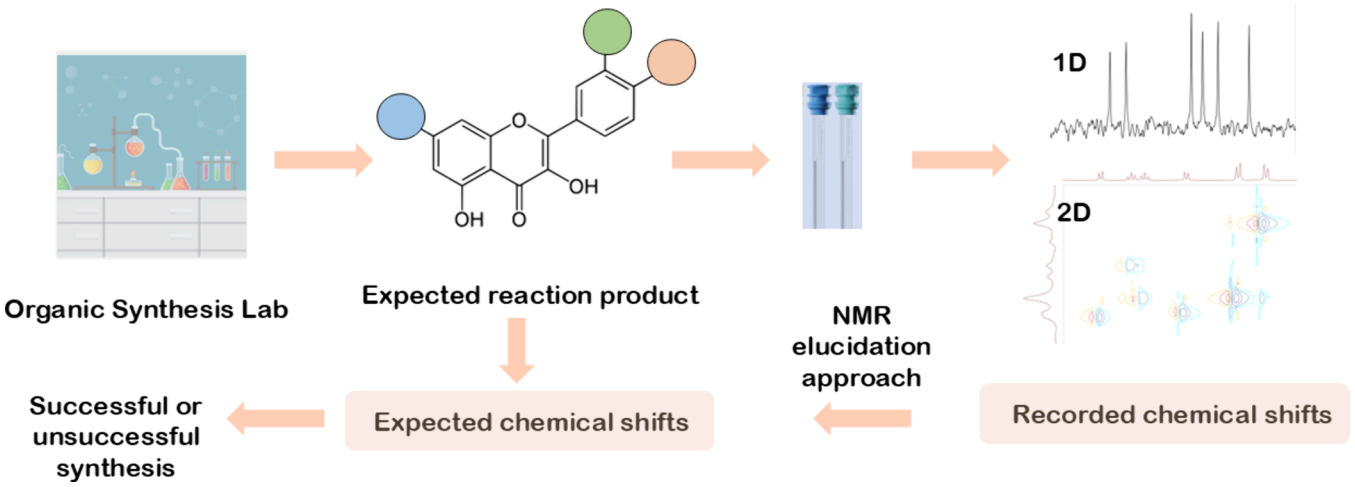


Figure 1. Proposed teaching approach.

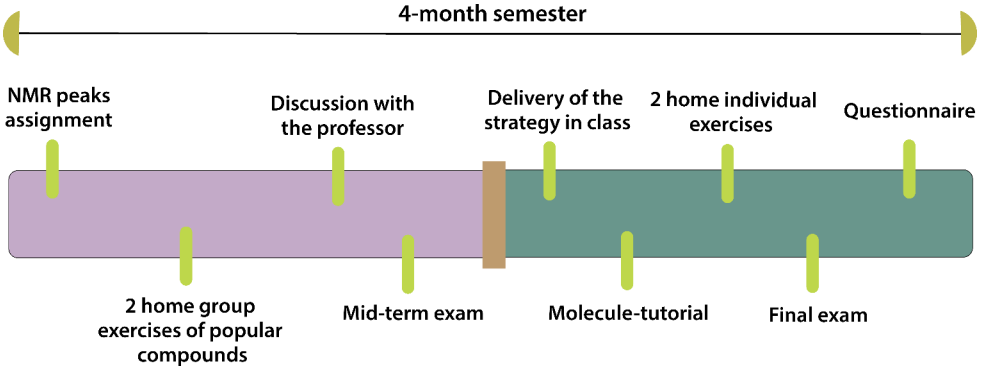


Figure 2. Time-dependent scheme depicting the curriculum of the class.

chemical shifts of the molecular structure. After performing an investigation and discussing with the students, we observed that their main difficulty is the simultaneous management of multiple spectra given during the exams. After conducting extended bibliographic research concerning the proposed strategies to treat NMR spectra, we discovered that many different strategies have been proposed for the identification of molecules' chemical structures. In the literature, several detailed examples are indicated; however, they usually refer to a particular category of compounds (e.g., flavonoids, saccharides),[1−8] compound substructures[9] or to the assignment of peaks of protons and carbons of a compound solely by using 1D NMR spectra.[10−14] Furthermore, significant techniques have been proposed to determine the chemical structures using a combination of methods, such as ^{1}H NMR and IR.[15] Hence, these examples do not establish a general approach for assigning chemical shifts in any given chemical structure through NMR spectroscopy.

To deliver a thorough approach concerning structure elucidation through the use of NMR spectroscopy and to assist undergraduate students in simultaneously managing 1D and 2D spectra, we developed a teaching approach that proved to be very helpful for students struggling with NMR structure elucidation. Ultimately, students enrolling in this course learn how to develop a strategy to interpret 1D and 2D NMR homonuclear and heteronuclear spectra for assigning the peaks of protons and carbons of any given molecule. This teaching approach focuses on helping students develop the ability to predict the chemical shifts of ^{1}H and ^{13}C NMR peaks and the corresponding homonuclear or heteronuclear correlations *a priori*, solely by analyzing a molecule's structure—without relying on the provided spectra. This teaching approach can be applied in two steps. The first step is based on the prediction of proton, carbon, or proton-carbon correlations only by observing a molecule's chemical structure and then listing all the signals expected to be present in all the available spectra. The second step focuses on the observation of the correlations and signals present in 1D and 2D spectra in comparison to those that have been already predicted. This two-step method has proven to be valuable for undergraduate students, since they do not need to struggle simultaneously with a number of spectra, but instead, they divide the long task of structural identification into separate easier and more organized sub jobs. Thus, the complicated task of assigning proton and carbon peaks is transformed to an effortless application of an organized theory (Figure 1).

To deliver this approach to the classroom, we organized a variety of activities (Figure 2). We designed the curriculum of the class to evaluate the teaching approach in terms of students' grades, performance in class and opinion on the proposed method. The students enrolled in this course are in the sixth semester of their studies in the Department of Chemistry, and they have been already taught the basic concepts of NMR in another course of the curriculum (5th semester); however, they have not faced a large number of practical examples in their studies. Thus, our educational plan

Downloaded from pubs.acs.org/jceda8/article-pdf/102/8/3226/4129173/ed4c00402.pdf by Dong Ziyu on 10 October 2026

https://doi.org/10.1021/acs.jchemed.4c00402
J. Chem. Educ. 2025, 102, 3226−3233

for the 4 month semester was divided into two periods. The first half of the semester included teaching the basics of NMR spectra interpretation (1D and 2D) without mentioning any proposed teaching approach as well as familiarizing the students with practical examples, while the second half of the semester was dedicated to the delivery of the teaching approach to the class and the students' acquaintance with it. Additionally, software tools were demonstrated in class to help students visualize molecular structures and predict valid NOE correlations based on 3D geometry. Programs such as Maestro Schrödinger,[16] Chimera,[17] and Pymol[18] were used to illustrate how proton spatial proximity can lead to observable NOE interactions.

During the first weeks of the semester, the students were assigned two home group exercises for structure elucidation of known molecules (drugs or natural products), such as curcumin and losartan. We chose to use popular compounds (i.e., aspirin, Δ^8-tetrahydrocannabinol, cocaine, TRF tripeptide, ibuprofen, morphine, azidothymidine, etc.) in order to make the class more appealing to the students and pinpoint the importance of having a "clean" spectrum for molecules that are daily consumed by humans. After the completion of home exercises, one discussion with the professor was organized per group to monitor the problems the students faced during the home exercise. Social learning and discussion acquisition enriches the logical and conceptual growth;[19] thus by organizing the discussion with the professor, we gained valuable feedback from the students, while the students could engage more with the educational process.

At this point of the semester, the midterm exam was organized. The students were given the necessary 1D and 2D homonuclear and heteronuclear spectra of a known compound, and their task was to identify the proton and carbon chemical shifts from all of the available spectra. The midterm exam was graded by the professor mainly for statistical reasons and contributed 20% to their final grade, while our intention was to evaluate the students based on the grade of the final exam, after having taught them the implementation of the approach, which contributed 80% to their final grade.

During the second half of the semester, the students were taught the proposed approach in class with two examples of known compounds that they had already interpreted in the first half of the semester. After having explained the approach in class, each student was assigned with one molecule tutorial (the same molecule was given to all the students along with guidelines on how to apply the new approach). Moreover, each student had to assign the proton and carbon peaks to two different molecules as a home exercise. The final exam followed the individual home exercises, where the students were asked to interpret the same spectra as in the midterm exam, but this time by applying the proposed approach. Finally, during the last class, the students were asked to evaluate the proposed approach with a questionnaire and to note all of the problems they faced while implementing it.

At this point, we present the proposed teaching approach for NMR structure elucidation. The following simple rules are applicable to organic compounds under study and are highly recommended:

(a) If there is an isolated methyl group in an alkyl chain, it is always a good starting point for the application of the teaching approach. The integration of the peak immediately determines the methyl group, and through 2D COSY correlations all the protons of the alkyl chain can be assigned unambiguously.

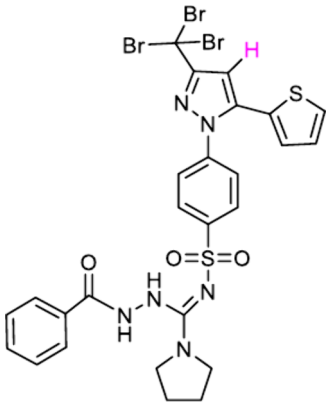
(b) 1,4 disubstituted aromatic compounds with two different substituents exhibit a doublet of doublets with peaks that are easily recognizable in 1H NMR spectra.

(c) 2D COSY and 2D NOESY can be used in a complementary fashion to fully assign the ^{1}H NMR spectrum.

(d) The factors that affect chemical shifts should also be considered during structural analysis (inductive effect, pH, resonance forms, solvent).

(e) The symmetry of the molecule should be examined carefully as it determines the number of the peaks presented in the spectra.

(f) Isolated single peaks can be easily recognized in the spectra. For example, the proton on the chemical structure illustrated below (proton highlighted in pink) will appear as a single peak in the aromatic region, as it does not couple with any other protons.



(g) It is particularly important to understand the complexity of the observed multiplet peaks. Students should keep in mind that homotopic and enantiotopic protons are not differentiated in NMR spectroscopy in contrast with diastereotopic ones.

(h) After assigning all protons of the structure, 2D HSQC can easily provide the assignment of the carbons that are directly bonded with protons.

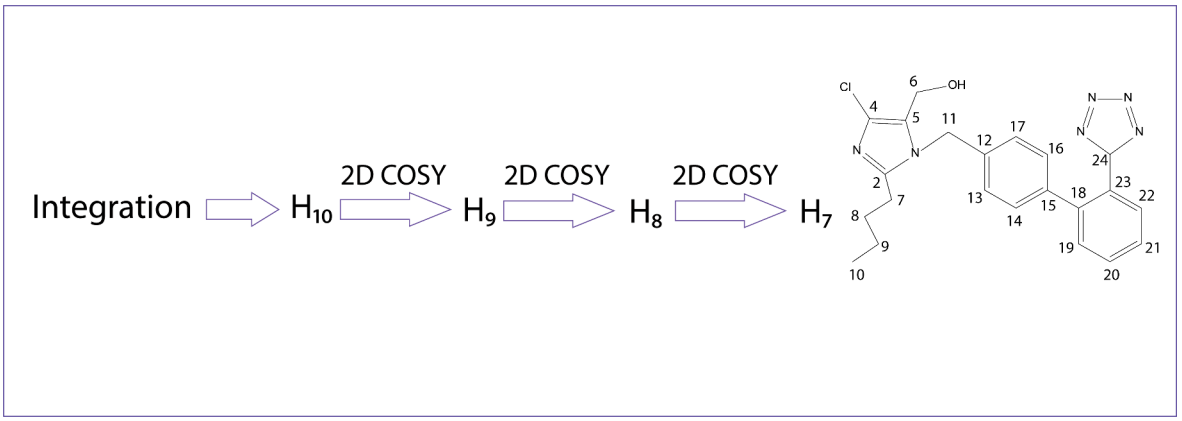
(i) The last step is the assignment of the tertiary carbons. Students should understand that the assignment of tertiary carbons is accomplished by using 2D HMBC spectra and by exploiting all proton-carbon correlations, revealing the atoms residing in vicinal proximity. They should also know that tertiary carbons usually resonate at low intensities.

This teaching approach following the aforementioned rules has been taught to students for many years and has presented promising results. After solving many examples using this approach during class, the students are required to find the spectra of a compound of their interest and test their own approach during the examination. When their developed approach does not match perfectly with the practical assignment, they realize that they could modify their theory. In this way, theory and practice complement each other. The NMR teaching approach has been found useful for undergraduate students conducting their thesis in the organic chemistry lab since they struggle with structure elucidation challenges on a daily basis. In particular, students need to

Downloaded from pubs.acs.org/jceda8/article-pdf/102/8/3226/4129173/ed4c00402.pdf by Dong Ziyu on 10 October 2026

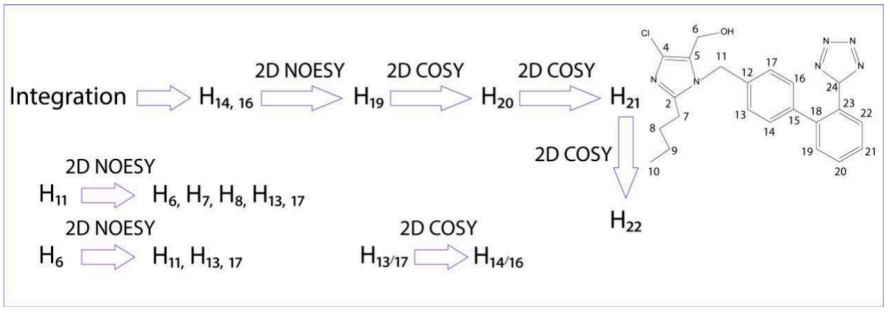
https://doi.org/10.1021/acs.jchemed.4c00402
J. Chem. Educ. 2025, 102, 3226−3233

Scheme 1. Designation of the Structure Elucidation of an Alkyl Chain[a]



[a]The teaching approach starts from H_{10} and through 2D COSY experiment can be assigned the protons H_9, H_8 and H_7.

Scheme 2. Designation of the Structure Elucidation of a Biphenyl Ring[a]



[a]The teaching approach starts from $H_{14,16}$ and protons H_{20}, H_{21} and H_{22} can be assigned through 2D COSY and 2D NOESY experiments. H_{11} can be also differentiated by H_6.

evaluate the reactions' products, by analyzing their samples using NMR. The proposed teaching approach relies on the behaviorist approach[20] that has been proved very reliable and effective in learning. It involves a step-by-step procedure based on set rules. Our approach focuses on mastering early steps before processing more complex levels of performance. The first two examples illustrated below exemplify this approach. In particular, losartan is an ideal molecule for students to start applying this methodology, while the second example, curcumin, is more difficult since the molecule exists in keto and enol forms. A steroid compound has also been included in the curriculum as an example of a complex molecule featuring cyclohexane substructures. A detailed application of the method is presented in the Supporting Information for the three examples, as well as for the fourth example, 9-hydroxyphenalenone, which has already been published in the Journal of Chemical Education and is included solely for comparison with the proposed approach (Supporting Information, Scheme S3).

DELEVOPMENT OF TEACHING APPROACH – LOSARTAN

Losartan is a commercially available pharmaceutical molecule prescribed for the treatment of hypertension. It belongs to the class of sartans and acts by antagonizing the detrimental effect of octapeptide hormone Angiotensin II (AII) to prevent its hypertensive effect in pathological states.[21,22] By acting as an AT1R antagonist, losartan prevents AII's binding to the AT1 receptor and induces the antihypertensive activity.[23] In Figures S1–S7, the homonuclear 1D NMR spectra and homonuclear (2D COSY and NOESY) and heteronuclear 2D NMR spectra (HSQC and HMBC) are illustrated.

According to this approach, it is always important to select a starting point. For this molecule, we notice the presence of a terminal methyl group, integrated into three protons and indicating one triplet peak, since it has only two adjacent protons. Through 2D COSY, all other protons of the butyl chain can be unambiguously assigned. The multiplicity and integration of the peaks can confirm the assignment. This process can be easily interpreted through arrow annotations as depicted in Scheme 1.

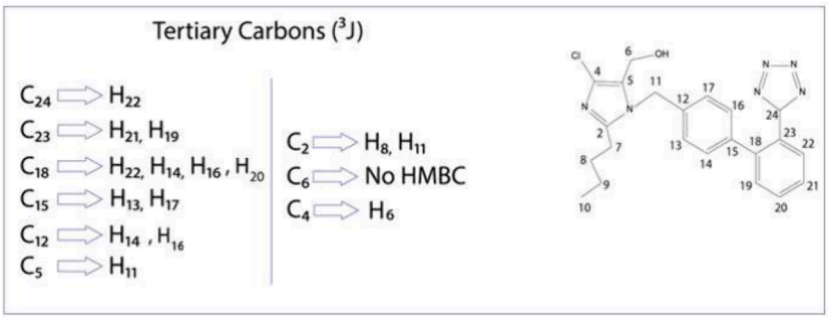
The remaining aliphatic peaks, H_6 and H_{11}, are assigned as a single peak (resonate at 4.3 and 5.5 ppm respectively). The differentiation between the aforementioned protons can be accomplished through the 2D NOESY spectrum since H_{11} protons indicate NOE correlations with H_{13}/H_{17} protons.

The second critical starting point is the NOE correlation between the H_{14}, H_{16} and H_{19}, which are protons in the spatial vicinity. $H_{14,16}$ presents a doublet peak whose integration reveals two protons. The phenyl ring that is connected with a methylene bridge presents two doublet peaks, verifying the expected pattern of the 1,4 disubstituted ring. Since NOE correlations between H_{14}, H_{16} and H_{19} can be identified through the 2D NOESY spectrum, the structure identification of the three protons is accomplished. Finally, by using 2D COSY spectra, all of the other protons can easily be identified (Scheme 2).

https://doi.org/10.1021/acs.jchemed.4c00402
J. Chem. Educ. 2025, 102, 3226−3233

Downloaded from pubs.acs.org/jceda8/article-pdf/102/8/3226/4129173/ed4c00402.pdf by Dong Ziyu on 10 October 2026

Scheme 3. Designation of the Structure Elucidation of Tertiary Carbons[s]



[s]As can be easily observed, tertiary carbons indicate different correlations that permit their unambiguous assignment. For example, C_2 and C_5 are differentiated, since C_5 presents one HMBC correlation with H_{11}, while C_2, shows one additional correlation with H_8.

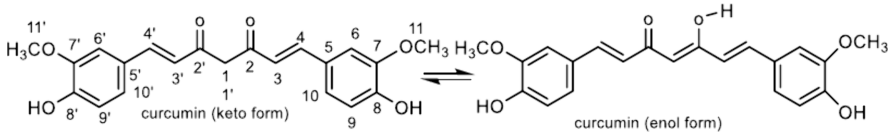


Figure 3. Keto- and enol-forms of curcumin.

The 2D HSQC spectrum permits assignment of all carbons bearing protons. Furthermore, 2D HMBC allows the complement of the assignment through the structure identification of tertiary carbons, as illustrated below in Scheme 3.

DELEVOPMENT OF TEACHING APPROACH – CURCUMIN

Curcumin is another molecule example that will be presented in order to illustrate the proposed identification approach. Curcumin is a symmetric natural compound, which is widely used in the cosmetics and food industries. Although the molecule is symmetric, it exists in two forms (keto-form and enol form),[24] where in enol form the symmetry is broken. Protons H_{2-11} and carbons C_{2-11} are equivalent to protons or carbons 2′-11′, as illustrated in the equilibrium below (Figure 3).

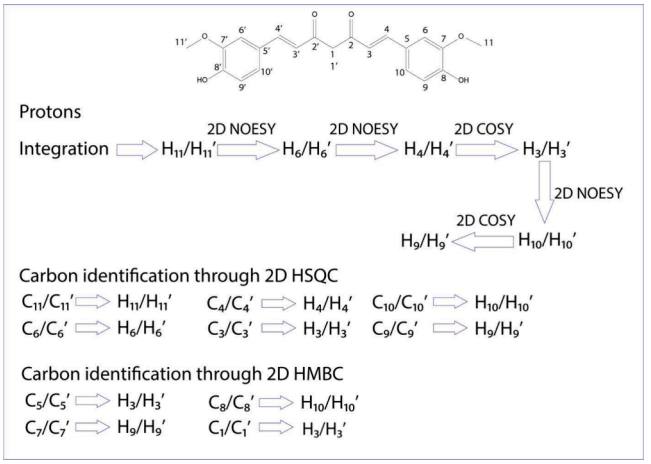
In this case, the interpretation of the spectra is a rigorous project demanding a lot of effort from the students, since they need to distinguish the peaks for the two different forms (keto form and enol form) in curcumin's equilibrium. Thus, we recommend to the students to identify, for practice, the molecule as if it exists only in its keto form, for the scope of this course. As a more advanced NMR expert, it is recommended that students analyze the whole spectra, taking into account both the enol and keto forms of the molecule. This is suggested to master's students or to students that conduct their undergraduate thesis in the organic synthesis lab. Students must be taught that peaks are more deshielded in the keto form than in the enol form. Spectra can be found in Figures S8–S13.

The approach that can be used for its structure elucidation is described in Scheme 4.

DELEVOPMENT OF TEACHING APPROACH – ORGANOTIN STEROID

Organotin steroids are emerging as potential anticancer agents for targeting breast cancer, particularly hormone-dependent types. These compounds combine the selective binding properties of steroids with the cytotoxic effects of organotin

Scheme 4. Teaching Approach for Proton and Carbon Identification of Curcumin Using All the Available Spectra (^{1}H, ^{13}C, 2D COSY, 2D NOESY, 2D HSQC, 2D HMBC). (1-1′)



moieties. Studies have shown that organotin derivatives can induce apoptosis in breast cancer cells by disrupting mitochondrial function and inhibiting key enzymes involved in tumor growth.[25]

This approach can be applied to the structure elucidation of an organotin derivative drug candidate for breast cancer treatment, as described in Scheme 5. Spectra can be found in Figures S14–S20.

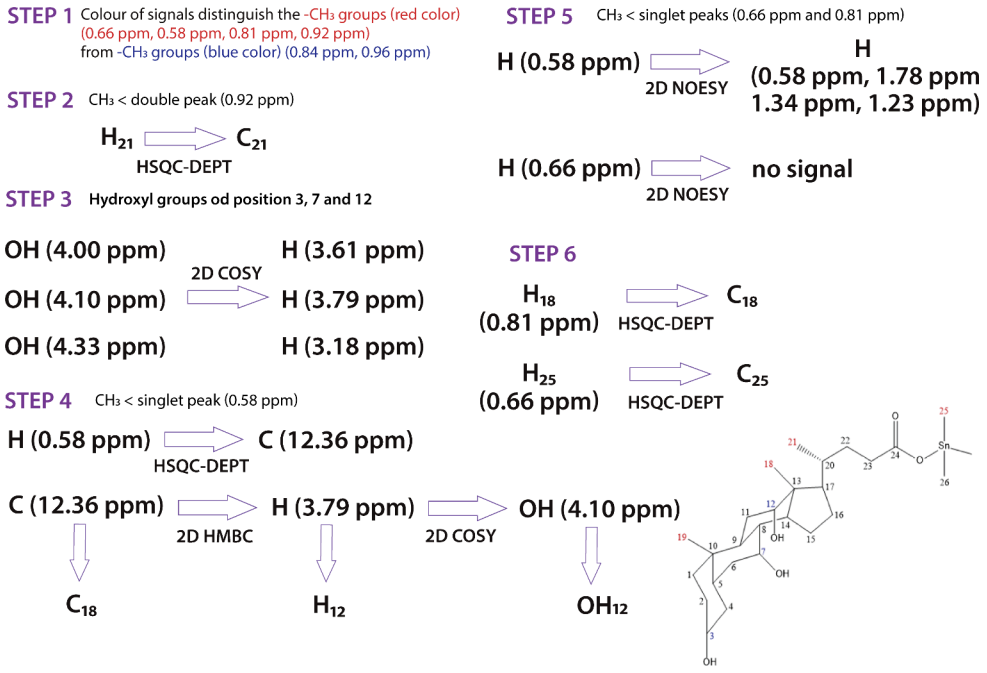
DEVELOPMENT OF TEACHING APPROACH – PEDAGOGICAL POINT OF VIEW

In the beginning of the semester, the curriculum of the class started with learning the basic principles of NMR spectroscopy, where the initial guidelines to interpret 1D and 2D NMR spectra were imparted. The students had already been taught the basic concepts of NMR spectroscopy in another course during the curriculum of the chemistry department (previous semester); thus, adding more examples and the concept of 2D

Downloaded from pubs.acs.org/jcesa8/article-pdf/102/8/3226/4129173/ed4c00402.pdf by Dong Ziyu on 10 October 2026

https://doi.org/10.1021/acs.jchemed.4c00402
J. Chem. Educ. 2025, 102, 3226–3233

Scheme 5. Teaching Approach for Proton and Carbon Identification of Organotin $Me_2Sn(CA)_2$ by Using All the Available Spectra (^1H, ^{13}C, 2D COSY, 2D NOESY, HSQC-DEPT, 2D HMBC)



NMR was quite successful. However, the home group exercises that were assigned to the students during the first weeks did not have the same fate.

As mentioned earlier in this paper, the classroom curriculum allowed for evaluation of the teaching approach in three parts: (i) students' grades, (ii) performance in class, and (iii) their opinion of the proposed method.

Concerning the students' grades in the final exam when compared to those of the midterm exam (during the spring semester of 2023), they clearly indicate that the performance of the students was improved after application of the approach. Each year, the midterm and final exams feature different molecules which are mainly selected from a pool of well-known compounds, including natural products and pharmaceuticals. The participants were 109 students in the optional course "NMR Spectroscopy and Molecular Modeling" at the National and Kapodistrian University of Athens. The attendees of this course are students in the sixth or eighth semester of their studies. A paired samples t test was conducted to compare the score differences before (midterm exam) and after (final exam) the application of the teaching approach. The results provided in Table 1 indicate that there was a significant difference ($t = -4.527$, df = 108, $p < 0.001$) in the students' grades before ($M = 8.35$, $SD = 1.60$) and after ($M = 9.11$, $SD = 1.18$) the implementation of the approach. These results suggest that the students succeeded with higher exam scores after they have been taught the proposed approach.

Table 1. Statistical Parameters of Students' Grades at Middle and Final Exams

Middle Exam grades (MEG)		Final Exam grades (FEG)				
Mean	Std. Deviation	Mean	Std. Deviation	t	df	Sig (2-tailed)
8.35	1.61	9.11	1.18	−4.527	108	<0.001[a]

[a]Significant at the level of $p < 0.05$.

Concerning the students' performance in class, their speed in performing the task in class was accelerated even right after the introduction of the approach. For the peak assignment of a natural compound, we needed 2 h of class before the explanation of the teaching approach, while after its explanation in the classroom, 75 min were enough. This was even more obvious after completing one more exercise (molecule-tutorial) based on the approach taught, where the students were able to perform the peak assignment faster and more easily.

Students' opinions concerning the curriculum of the class were asked in two phases. As reported to the professor, during the discussion with each group in the middle of the semester (phase A), the students faced several issues with trying to solve the home exercises in comparison to those explained in class, where they had the guidance of the professor. The main difficulties they faced were (i) spending a lot of time to discover a suitable starting point (45%), (ii) the simultaneous manipulation of all the spectra (29%) and (iii) inability to complete the work even if they had managed to organize the spectra and perform a do-it-yourself (DIY) approach. The main positive points of the teaching that emerged in this phase were (i) the working in groups and (ii) the dealing with everyday compounds. Nevertheless, by working on the group exercises, students had the chance to exchange their ideas on the project, elaborate their own personal strategies to interpret NMR spectra, and perform the chemical shift assignments. Furthermore, the majority of the students highlighted that working on their project with others aided them in understanding the task and organizing their thoughts and spectra. Moreover, during the discussion with the professor, almost all of the students reported that it was quite interesting to identify the NMR peaks of known compounds that they consume daily. It kept them focused on the project and gave

Downloaded from pubs.acs.org/jceda8/article-pdf/102/8/3226/4129173/ed4c00402.pdf by Dong Ziyu on 10 October 2026

https://doi.org/10.1021/acs.jchemed.4c00402
J. Chem. Educ. 2025, 102, 3226−3233

them an idea of the kind of work that is performed in an industrial analytical laboratory.

Finally, according to the responses of the students in the final questionnaire (phase B), several benefits of implementing this step-by-step strategy were reported. Students assigned the peaks of protons and carbons of the molecules that were assigned to them as home exercises during the second half of the semester much more easily than during the first half. Furthermore, they indicated that the approach (as well as the use of arrow notation) aided them in organizing their thoughts and in managing multiple spectra. Most of them reported that the process of peak assignment became clearer to them and that the applications of these rules helped them to create an outline of the structure elucidation process. They also indicated that it took them less time to finish the home exercises by implementing the proposed approach, even if the tasks were assigned individually rather than as a group project. Interestingly, a few students reported that they had already created similar strategies in their minds to organize their thoughts on the management of spectra and assign proton and carbon NMR peaks. These students compared this task with jigsaw puzzle solving. The benefits from the application of the teaching approach are illustrated in Scheme 6.

Scheme 6. Benefits from the Application of the Strategy



Furthermore, we need to highlight that this course is resolute solely in teaching 1D and 2D NMR Spectroscopy and does not include any supplementary analytical techniques. Thus, the structure assignment proposed in this approach can be based solely on NMR spectroscopy. However, in the case of an extract from a natural product, supplementary techniques, such as HPLC, MS, and IR, should be used for structure identification. Moreover, our approach is useful for those students studying organic or medicinal chemistry who in their daily practice know the compounds they are trying to synthesize and can use the approach to confirm or reject the compounds obtained.

CONCLUSIONS

The development of a teaching approach for structure elucidation through homonuclear and heteronuclear NMR spectroscopy is a necessary tool for both undergraduate and graduate students working in a variety of chemical laboratories. By using this approach, students have the chance to apply all their knowledge regarding integration and the factors affecting the chemical shift, as well as the homonuclear and heteronuclear correlations. Furthermore, they can justify their own applied approach based on the experimental results and quickly and effortlessly interpret the important spectra in each case to elucidate the desired structure. This teaching approach is beneficial for students because it primarily relies on theory. First, students predict all the possible expected NMR peaks by analyzing the molecule's structure. Then, they compared these predictions with the corresponding peaks in the spectra to check if they align with their expectations.

Moreover, by including all these learning activities (home exercises, group exercises) in the curriculum of the class, we accomplished the students' engagement with the method and their total interaction with the theory.

This teaching approach is used as the primary methodology in an optional course designed for students conducting their undergraduate thesis in an organic or medicinal chemistry lab. In this setting, students are regularly required to identify expected reaction products, unknown structures, impurities, and byproducts using NMR. As a result, they already have a chemical structure in mind and interpret the NMR spectra to confirm their predictions.

Hence, this teaching approach is very effective for them to confirm or reject the synthesis of the expected product. This strategy is effective because it relies on a method of thinking that students are already familiar with, which is to start from the implementation of the theory and then proceed to the interpretation of experimental results and not *vice versa*. Finally, this teaching approach could not be applied for unknown chemical structures, as additional analytical methods are required to accomplish molecule identification.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available at https://pubs.acs.org/doi/10.1021/acs.jchemed.4c00402.

Figures S1−S13. ^{1}H, ^{13}C, 2D COSY, 2D NOESY, 2D HSQC, 2D HMBC spectra recorded for losartan in D_2O at 27 °C and for curcumin in $CDCl_3$ at 27 °C. Figures S14−S20. ^{1}H, ^{13}C, HSQC-DEPT, 2D HMBC, 2D COSY, 2D NOESY spectra recorded for $Me_2Sn(CA)_2$ in DMSO-d_6 at 27 °C. Schemes S1 and S2. Overall schemes of the teaching approach. Schemes S3 and S4. Teaching approaches for proton and carbon identification of 9-hydroxyphenalenone and $Me_2Sn(CA)_2$ by using all the available spectra (^{1}H, ^{13}C, 2D COSY, 2D NOESY, 2D HSQC, 2D HMBC). Discussion and explanations of the teaching approaches (PDF)

AUTHOR INFORMATION

Corresponding Author

Thomas Mavromoustakos − *Department of Chemistry, National and Kapodistrian University of Athens, 15772 Athens, Greece*; Email: tmavrom@chem.uoa.gr

Authors

Eleni Chontzopoulou − *Department of Chemistry, National and Kapodistrian University of Athens, 15772 Athens, Greece*; orcid.org/0000-0002-3473-3312

Andromachi Tzani − *Laboratory of Organic Chemistry, School of Chemical Engineering, National Technical University of Athens, 15780 Athens, Greece*

Katerina Paschalidou − *Department of Chemistry, National and Kapodistrian University of Athens, 15772 Athens, Greece*; orcid.org/0000-0003-1938-5134

N. Zoupanou − *Department of Chemistry, National and Kapodistrian University of Athens, 15772 Athens, Greece*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.jchemed.4c00402

Downloaded from pubs.acs.org/jcedu8/article-pdf/102/8/3226/4129173/ed4c00402.pdf by Dong Ziyu on 10 October 2026

Funding

The open access publishing of this article is financially supported by HEAL-Link.

Notes

The authors declare no competing financial interest.

Ethical Statement. This study was conducted under the general institutional protocol for classroom activities at the National and Kapodistrian University of Athens. All participants provided informed consent, and data were anonymized for privacy protection. No safety hazards were encountered in this study.

■ ACKNOWLEDGMENTS

Materials were supported by Special Account for Research Grants (SARG), National Kapodistrian University of Athens (NKUA). Authors would like to acknowledge all the undergraduate and graduate students who attended the course of "Spectroscopy of Organic Compounds and Molecular Modelling" and evaluated the suggested NMR strategy.

■ REFERENCES

(1) Bross-Walch, N.; Kühn, T.; Moskau, D.; Zerbe, O. Strategies and Tools for Structure Determination of Natural Products Using Modern Methods of NMR Spectroscopy. *Chem. Biodivers.* **2005**, *2* (2), 147−177.

(2) Brown, G. D.; Bauer, J.; Osborn, H. M. I.; Kuemmerle, R. A Solution NMR Approach to Determine the Chemical Structures of Carbohydrates Using the Hydroxyl Groups as Starting Points. *ACS Omega* **2018**, *3* (12), 17957−17975.

(3) Caes, B.; Jensen, D. Synthesis and Characterization of 9-Hydroxyphenalenone Using 2D NMR Techniques. *J. Chem. Educ.* **2008**, *85* (3), 413.

(4) Hernández-Bolio, G. I.; Peña-Rodríguez, L. M. NMR Identification of Biologically Active Natural Products: Strategies and Challenges. *Natural Products as Source of Molecules with Therapeutic Potential* **2018**, 465−494.

(5) Kwan, E. E.; Huang, S. G. Structural Elucidation with NMR Spectroscopy: Practical Strategies for Organic Chemists. *Eur. J. Org. Chem.* **2008**, *2008* (16), 2671−2688.

(6) Zhang, Y.; Yamaguchi, T.; Kato, K. New NMR Tools for Characterizing the Dynamic Conformations and Interactions of Oligosaccharides. *Chem. Lett.* **2013**, *42* (12), 1455−1462.

(7) Gehman, C. A.; Hantz, E. R.; Hamann, C. S. Distinguishing Vinylic and Aromatic 1H NMR Signals Using Selectively Deuterated Chalcones. *J. Chem. Educ.* **2021**, *98* (7), 2323−2332.

(8) Dos Anjos, R.; Silva, M. S. Introduction Biodiesel Course: Evaluating the Quality of Waste Cooking Oil by 1H NMR Spectroscopy. *J. Chem. Educ.* **2020**, *97* (10), 3784−3790.

(9) Sanders, Jeremy K. M.; Hunter, B. K. *Modern NMR Spectroscopy: A Guide for Chemists*, 2nd ed.; Oxford University Press: Oxford, 1993.

(10) Topczewski, J. J.; Topczewski, A. M.; Tang, H.; Kendhammer, L. K.; Pienta, N. J. NMR Spectra through the Eyes of a Student: Eye Tracking Applied to NMR Items. *J. Chem. Educ.* **2017**, *94* (1), 29−37.

(11) Emwas, A.; Szczepski, K.; Poulson, B. G.; Chandra, K.; Mckay, R. T.; Dhahri, M.; Alahmari, F.; Jaremko, L.; Lachowicz, J. I.; Jaremko, M. NMR as a "Gold Standard" Method in Drug Design and Discovery. *Molecules* **2020**, *25* (20), 4597.

(12) Kolonko, E. M.; Kolonko, K. J. Introducing NMR Spectroscopy Using Guided Inquiry and Partial Structure Templating. *J. Chem. Educ.* **2019**, *96* (5), 912−919.

(13) Smith, K. T.; Hamann, C. S. Students Constructing for Themselves the Concept of Chemical Shift Correlation for Organic Substructures. *J. Chem. Educ.* **2024**, *101*, 223.

(14) Angawi, R. F. Using a Problem Solving-Cooperative Learning Approach to Improve Students' Skills for Interpreting 1H NMR Spectra of Unknown Compounds in an Organic Spectroscopy Course. *J. Chem. Educ.* **2014**, *91* (6), 823−829.

(15) Livengood, K.; Lewallen, D. W.; Leatherman, J.; Maxwell, J. L. The Use and Evaluation of Scaffolding, Student Centered-Learning, Behaviorism, and Constructivism to Teach Nuclear Magnetic Resonance and IR Spectroscopy in a Two-Semester Organic Chemistry Course. *J. Chem. Educ.* **2012**, *89* (8), 1001−1006.

(16) Schrödinger. Maestro | Schrödinger. *Schrödinger Release 2018−2*. **2018**.

(17) Ragno, R.; Tramontano, A.; Martinelli, A.; Tuccinardi, T. Molecular Docking Tutorial. *VI Eur. Work. Drug Des.* **2007**, 48.

(18) *The PyMOL Molecular Graphics System*; Schrödinger LLC, https://www.pymol.org/, 2015.

(19) Time, S. R.; Matching, S. S. *Encyclopedia of the Sciences of Learning*; Springer: Boston, 2012; DOI: 10.1007/978-1-4419-1428-6.

(20) Ertmer, P. A.; Newby, T. J. Behaviorism, Cognitivism, Constructivism: Comparing Critical Features From an Instructional Design Perspective. *Perform. Improv. Q.* **1993**, *6* (4), 50−72.

(21) Chontzopoulou, E.; Tzakos, A. G.; Mavromoustakos, T. On the Rational Drug Design for Hypertension through NMR Spectroscopy. *Molecules* **2021**, *26* (1), 12.

(22) Georgiou, N.; Chontzopoulou, E.; Routsi, E. A.; Stavrakaki, I. G.; Petsas, E.; Zoupanou, N.; Kakava, M. G.; Tzeli, D.; Mavromoustakos, T.; Kiriakidi, S. Exploring Hypertension: The Role of AT1 Receptors, Sartans, and Lipid Bilayers. *ACS Omega* **2024**, *9*, 44876.

(23) Zervou, M.; Cournia, Z.; Potamitis, C.; Patargias, G.; Durdagi, S.; Grdadolnik, S. G.; Mavromoustakos, T. Insights into the Molecular Basis of Action of the AT1 Antagonist Losartan Using a Combined NMR Spectroscopy and Computational Approach. *Biochim. Biophys. Acta - Biomembr.* **2014**, *1838* (3), 1031−1046.

(24) Manolova, Y.; Deneva, V.; Antonov, L.; Drakalska, E.; Momekova, D.; Lambov, N. The Effect of the Water on the Curcumin Tautomerism: A Quantitative Approach. *Spectrochim. Acta - Part A Mol. Biomol. Spectrosc.* **2014**, *132*, 815−820.

(25) Stathopoulou, M. E. K.; Zoupanou, N.; Banti, C. N.; Douvalis, A. P.; Papachristodoulou, C.; Marousis, K. D.; Spyroulias, G. A.; Mavromoustakos, T.; Hadjikakou, S. K. Organotin Derivatives of Cholic Acid Induce Apoptosis into Breast Cancer Cells and Interfere with Mitochondrion; Synthesis, Characterization and Biological Evaluation. *Steroids* **2021**, *167* (September2020), 108798.

Downloaded from pubs.acs.org/jceda8/article-pdf/102/8/3226/4129173/ed4c00402.pdf by Dong Ziyu on 10 October 2026